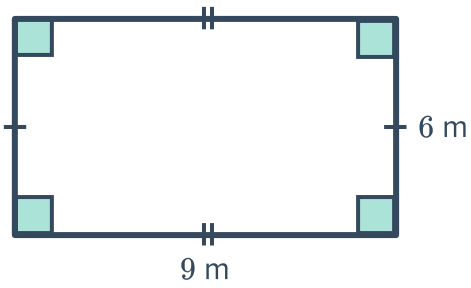
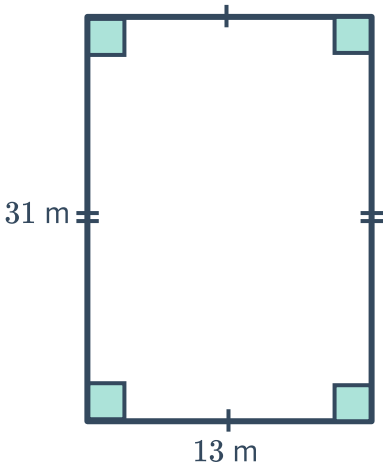
# Areas of rectangles, squares and triangles

**1** Find the area of the following rectangles:

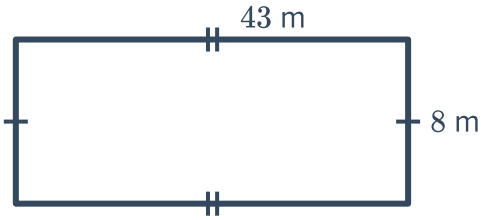
**a**



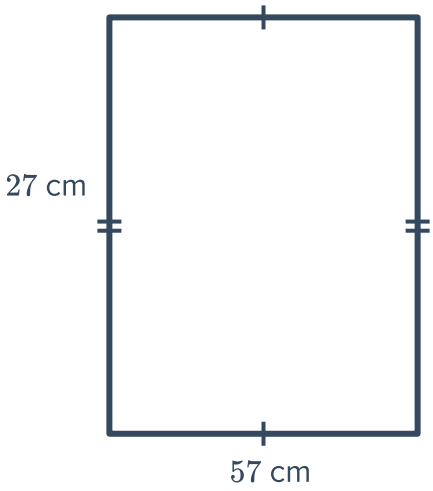

**b**




**c**




**d**




**2** Find the area of a rectangle whose length is $12\ \text{cm}$ and width is $5\ \text{cm}$.

**3** Find the area of a rectangle whose length is $11\ \text{m}$ and width is $8\ \text{m}$.

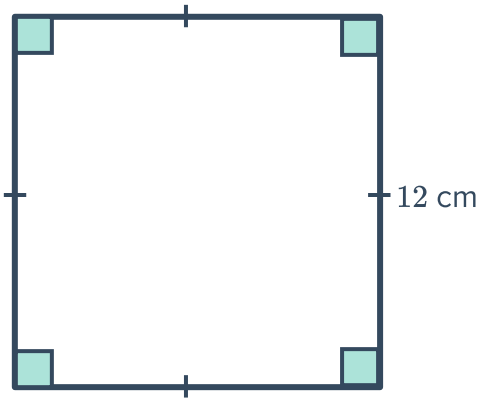
**4** If the area of a rectangle is $99\ \text{cm}^2$ and its width is $9\ \text{cm}$, find its length.

**5** If the area of a rectangle is $160\ \text{cm}^2$ and its length is $20\ \text{cm}$, find its width.

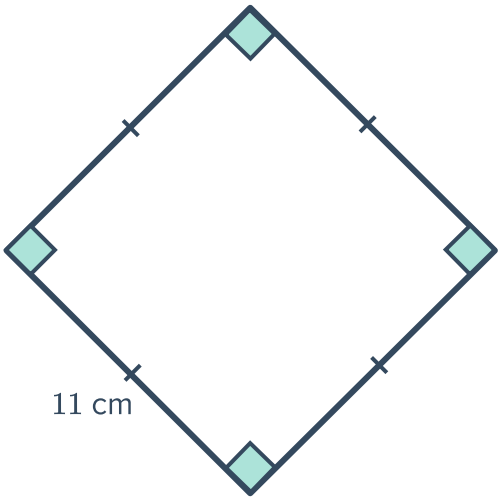
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**6** Find the area of the following squares:

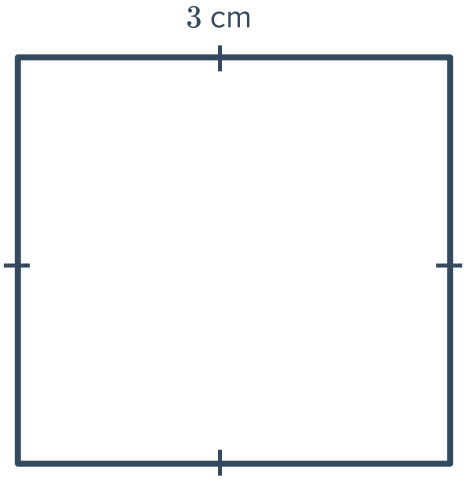
**a**




**b**




**c**




**7** Find the length of each side of a square whose area is:

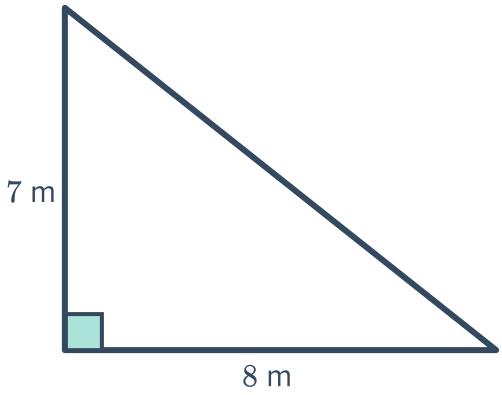
**a** $64\ \text{cm}^2$

**b** $36\ \text{m}^2$

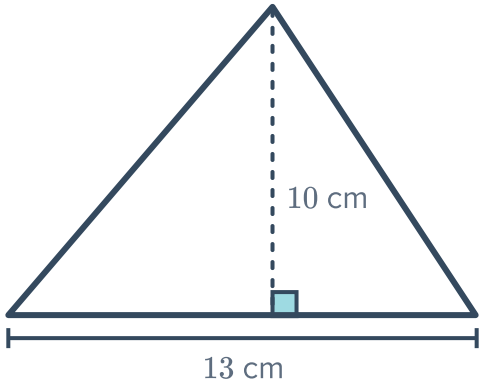
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**8** Find the area of the following triangles:

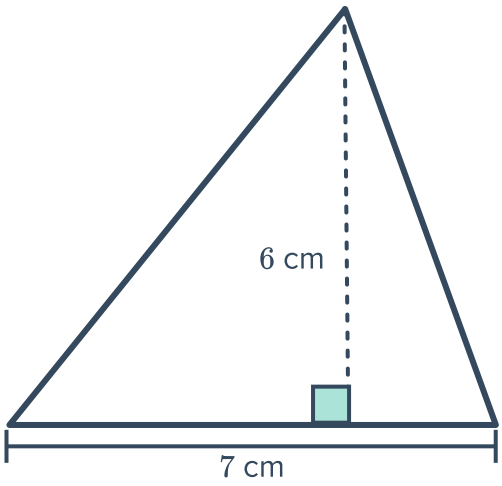
**a**



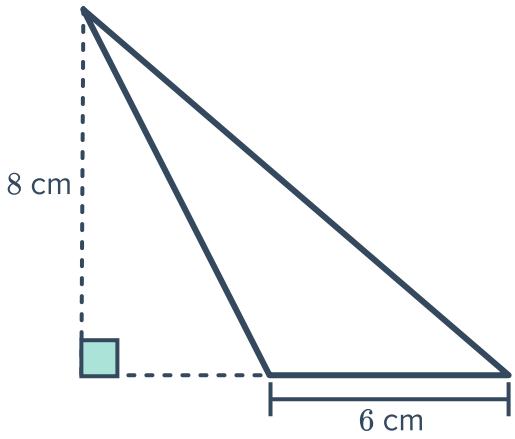

**b**



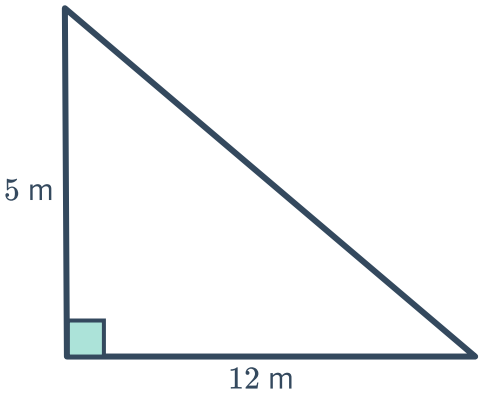

**c**



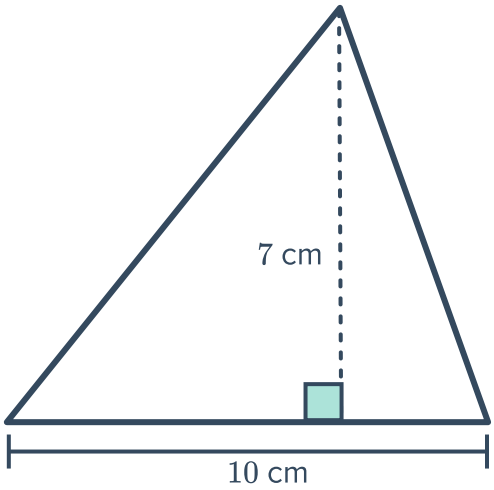

**d**




**e**

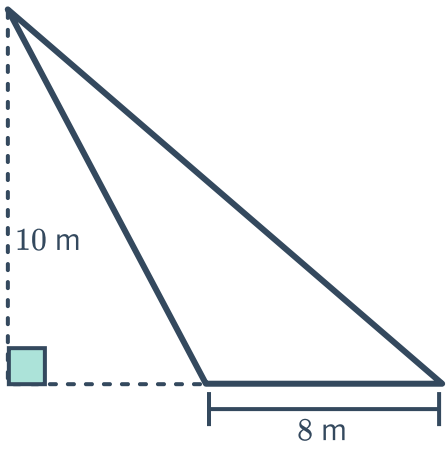



**f**



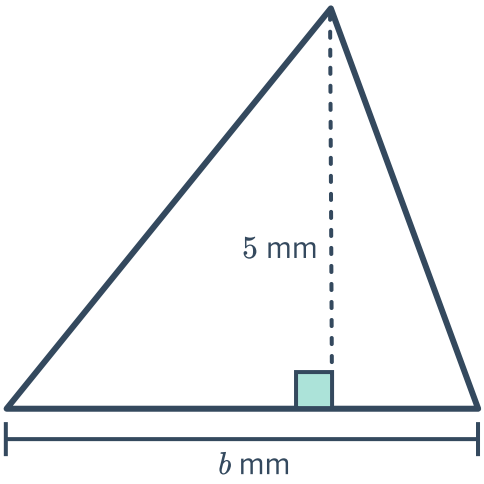

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**g**

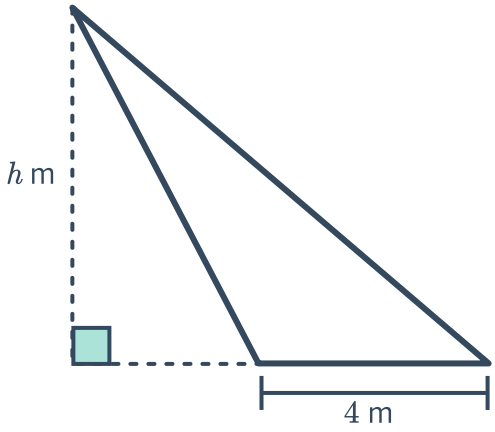



**9** Find the value of the pronumeral for each of the following triangles, given the area:

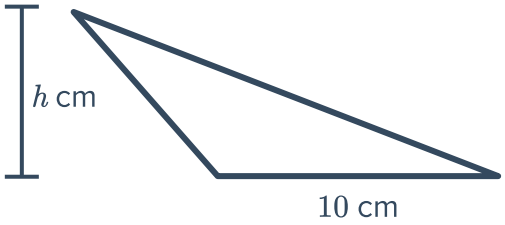
**a** The area of this triangle is $20\ \text{mm}^2$




**b** The area of this triangle is $48\ \text{m}^2$ .




**c** The area of this triangle is $130\ \text{cm}^2$.

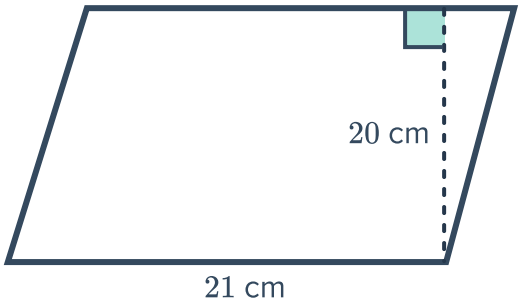



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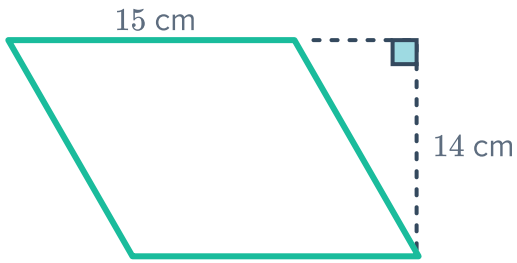
## Areas of parallelograms

**11** Find the area of the following parallelograms:

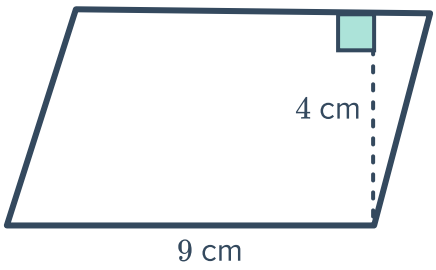
**a**



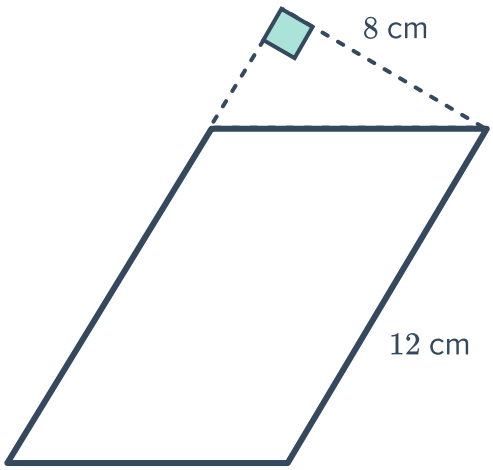

**b**




**c**




**d**




**12** Find the area of a parallelogram with:

- **a** A base of $5\ \text{mm}$ and perpendicular height of $2\ \text{mm}$.
- **b** A base of $15\ \text{cm}$ and a perpendicular height of $7\ \text{cm}$.
- **c** A base of $10\ \text{cm}$ and a perpendicular height $5\ \text{cm}$.

**13** Find the base length of a parallelogram with area of $96\ \text{cm}^2$ and perpendicular height of $8\ \text{cm}$ .

**14** Find the height of a parallelogram with area of $84\ \text{cm}^2$ and perpendicular base length of $6\ \text{cm}$

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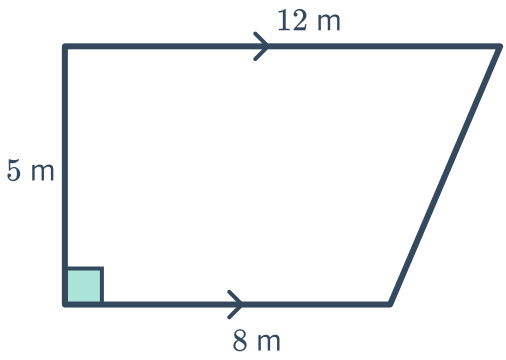
**15** Find the perpendicular height, $h$, of a parallelogram that has an area of $45\,\text{cm}^2$ and a base length of $5\,\text{cm}$.

**16** Find the base length, $b$, of a parallelogram that has an area of $216\,\text{mm}^2$ and a perpendicular height of $12\,\text{mm}$ .

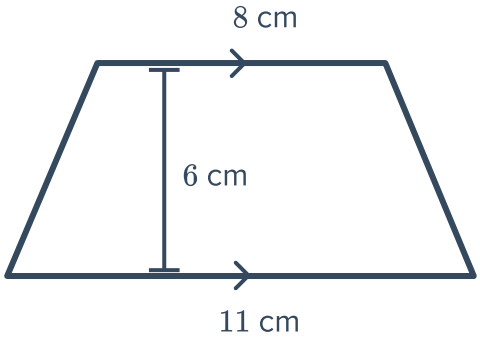
## Areas of trapeziums

**17** Find the area of the following trapeziums:

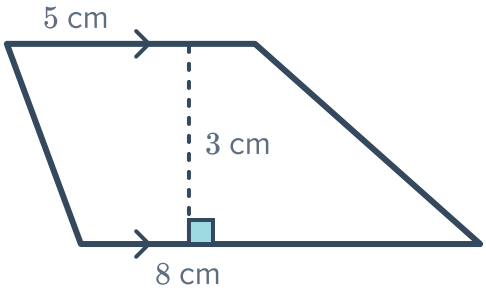
**a**



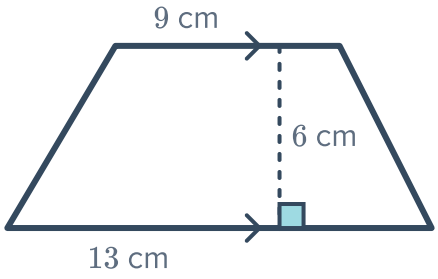

**b**




**c**

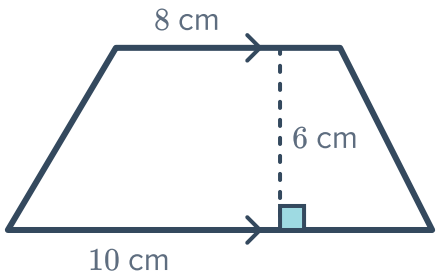



**d**

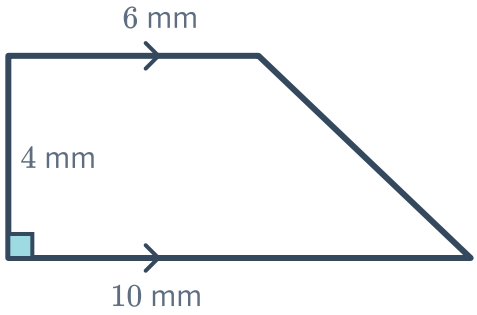



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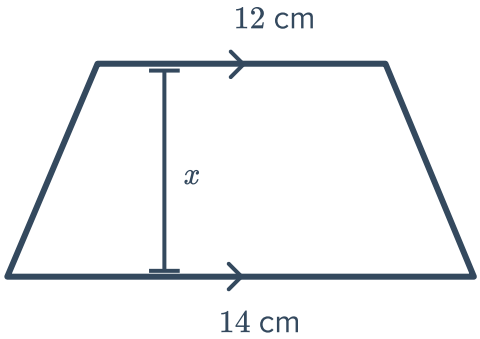
e



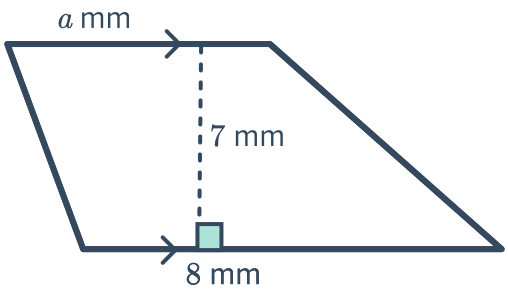

f




**18** Find the value of $x$ if the area of the trapezium shown is $65\ \text{cm}^2$.




**19** Find the value of $a$ if the area of the trapezium is $42\ \text{mm}^2$.

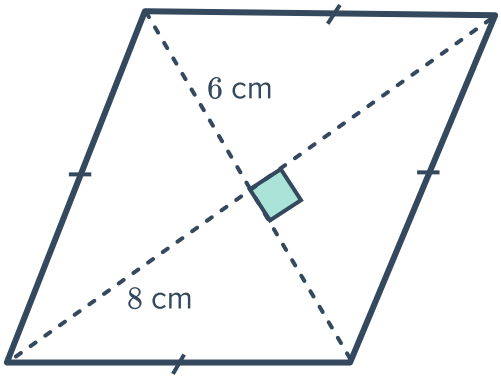



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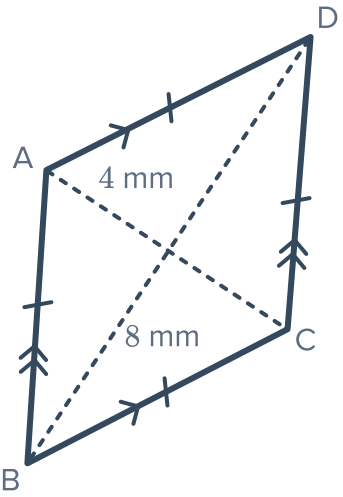
## Areas of Kites and Rhombuses

**20** Find the area of each rhombus below:

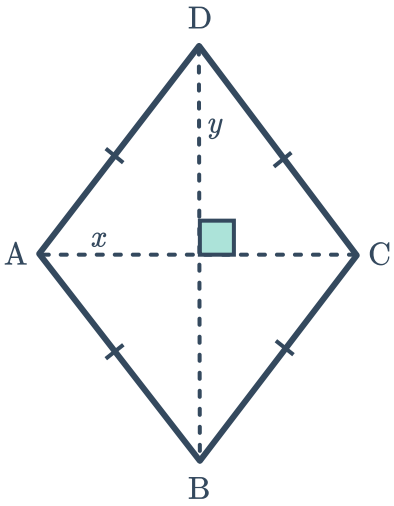
**a**



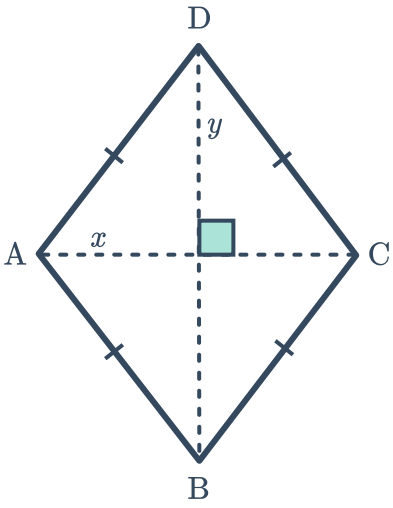

**b**



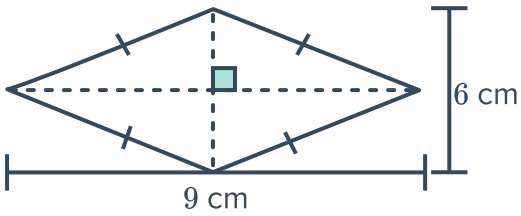

**c** $x = 10$ and $y = 12$




**d** $x = 4\,\text{cm}$ and $y = 8\,\text{cm}$

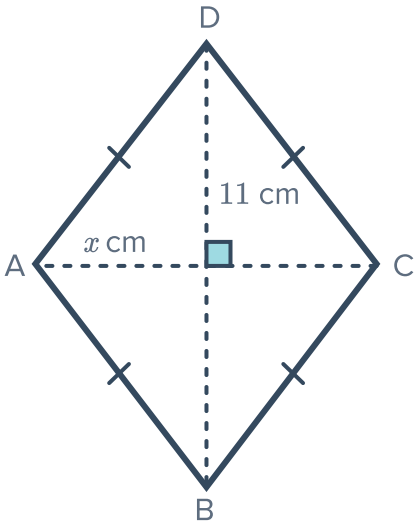



**e**



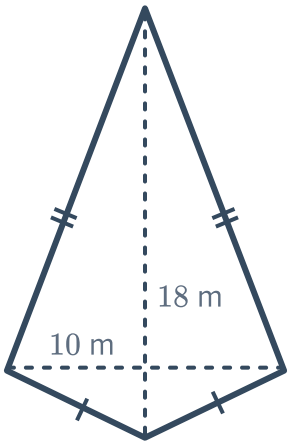

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**21** Rhombus $ABCD$ has an area of $55\,\text{cm}^2$. If diagonal $BD = 11\,\text{cm}$, and $AC = x\,\text{cm}$, find $x$.

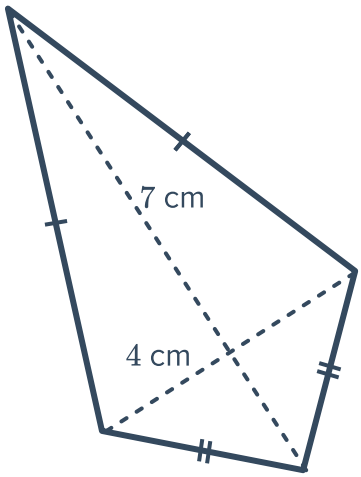



**22** Find the area of the following kites:

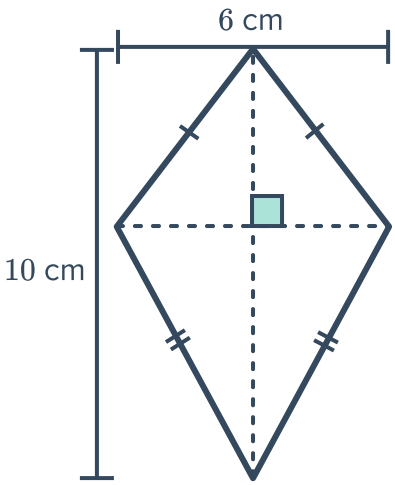
**a**




**b**



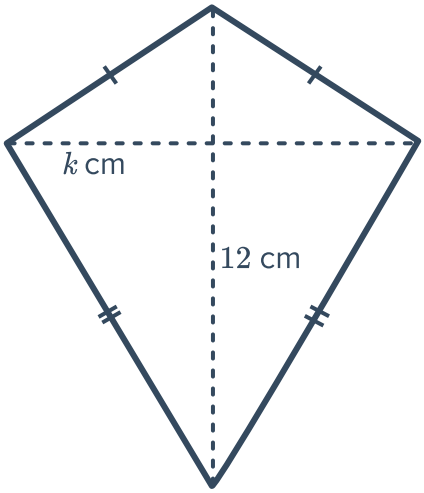

**c**




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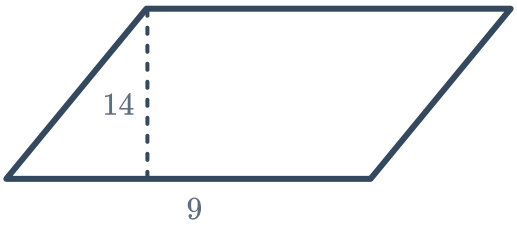
**23** The area of a kite is $308\ \text{cm}^2$ and one of the diagonals is $47$ cm. If the length of the other diagonal is $y$ cm, what is the exact value of $y$?

**24** The following kite has an area of $48\,\text{cm}^2$. The length of one of its diagonals is $12\,\text{cm}$. Find the length of the other diagonal, $k$.



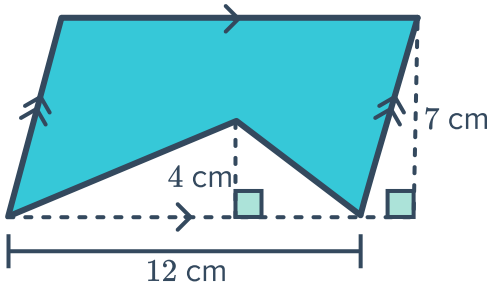

## Applications

**25** A rectangular sign casts a shadow on the ground as shown. What is the area of the shadow?



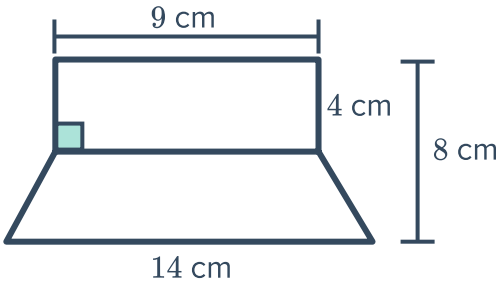

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**26** Find the shaded area shown in the figure:

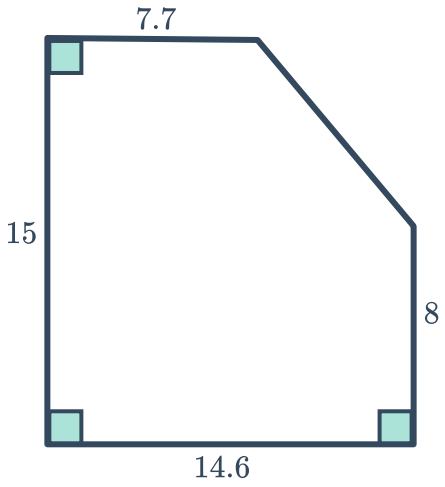



**27** Sharon has purchased a rectangular piece of fabric measuring 12 m in length and 7 m in width. What is the area of the largest triangular piece she can cut out from it?

**28** A rectangle is placed on top of a trapezium as shown. Find the area of the composite shape.

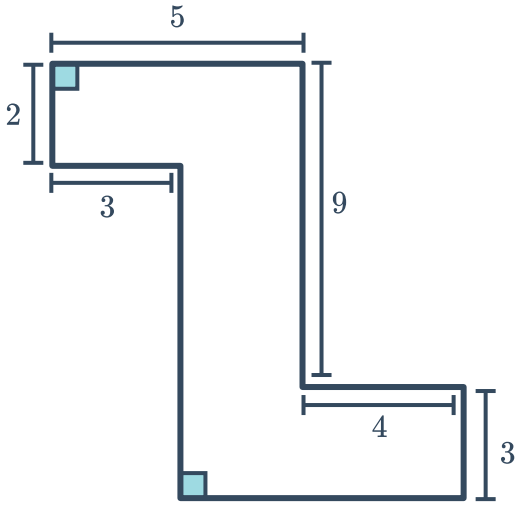



**29** Find the area of the figure shown. All dimensions are in millimetres.




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dimensions are in metres.




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